

ECSE 4000
Lec 6
Jan. 30

Topic: Conditional Probability
Total Probability Theorem
Bayes Rule

□ Example on Rolling two 6-sided Dice (Cont)

$$P(\underbrace{\text{Sum} \geq 8}_A \mid \underbrace{\text{One die} \geq 4}_B)$$

Definition

$$\downarrow \\ = \frac{P(A \cap B)}{P(B)}$$

$$P(B) = P(\text{One die} \geq 4) = 1 - P(B^c)$$

$$= 1 - P(\text{no die} \geq 4)$$

$$= 1 - P(\text{one die} < 4) \cdot P(\text{2nd die} < 4)$$

$$= 1 - \frac{1}{2} \times \frac{1}{2} = \frac{3}{4}$$

$$\begin{aligned} P(A \cap B) &= P(A) = P(\text{sum}=8) + P(\text{sum}=9) + P(\text{sum}=10) \\ &\quad + P(\text{sum}=11) + P(\text{sum}=12) \\ &= \frac{15}{36} = \frac{5}{12} \end{aligned}$$

$$P(\underbrace{\text{Sum is } \geq 8}_A \mid \underbrace{\text{one die} = 1}_B)$$

$$= \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

This example is for conditional probability in discrete sample space.

Similar ideas also hold for continuous sample space.

□ Example

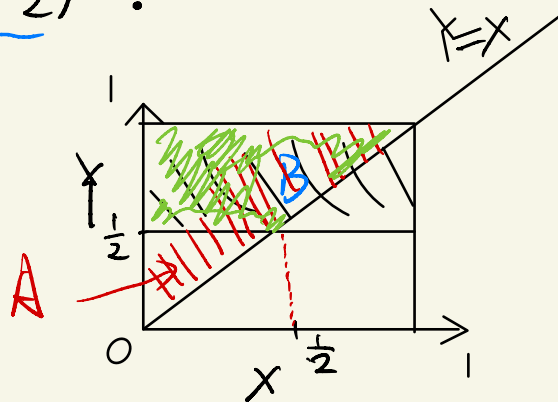
Two number (X, Y) uniformly randomly drawn from $[0, 1]$.

$$P(\underbrace{Y > X}_A \mid \underbrace{Y > \frac{1}{2}}_B) ?$$

$$= \frac{P(A \cap B)}{P(B)}$$

$$= \frac{\frac{1}{2} - \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}}{\frac{1}{2}}$$

$$= \frac{3}{4}$$

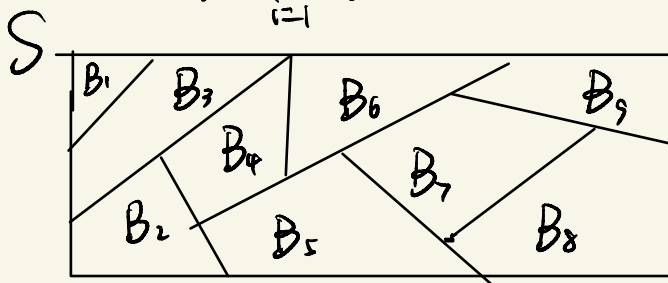


□ Now we introduce the idea of **a partition of the sample space S** :

- A set of Events $\{B_1, B_2, \dots, B_n\}$ such that

$$1) B_i \cap B_j = \emptyset, \quad \forall i \neq j = 1, 2, \dots, n$$

$$2) \bigcup_{i=1}^n B_i = S$$

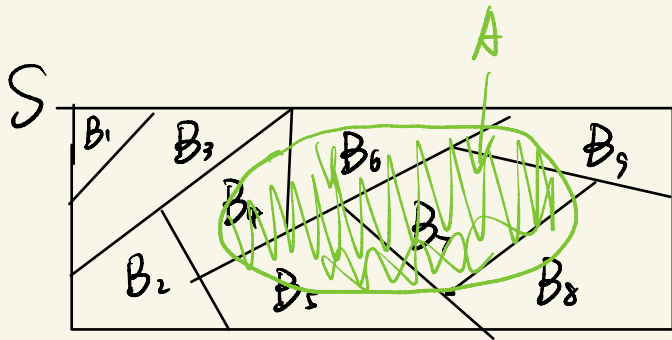


Based on Axiom 3.

Axiom 2

$$\sum_{i=1}^n P(B_i) = P\left(\bigcup_{i=1}^n B_i\right) = P(S) = 1$$

Now consider an Event A



$$\begin{aligned} P(A) &= P(A \cap S) \\ &= P(A \cap \left(\bigcup_{i=1}^n B_i\right)) \\ &= P\left(\bigcup_{i=1}^n (A \cap B_i)\right) \\ &= \sum_{i=1}^n P(A \cap B_i) \end{aligned}$$

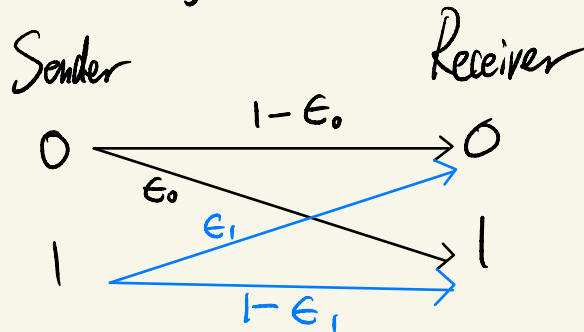
$$\Rightarrow P(A) = \sum_{i=1}^n \underbrace{P(A \cap B_i)} \\ = \sum_{i=1}^n P(A|B_i) P(B_i)$$

This is the so-called **Total Probability Theorem**.

This is useful since it relates probability of an event to its conditional probabilities with a set of possible events.

I will give several examples to show calculating $P(A)$ is difficult but computing $P(A|B_i) P(B_i)$ is simpler.

Example Binary communication channel



ϵ_0, ϵ_1 are
predefined constants.

$$P(\text{Rec: } 0 \mid \text{Send } 0) = 1 - \epsilon_0$$

$$P(\text{Rec: } 1 \mid \text{Send } 0) = \epsilon_0$$

$$P(\text{Rec: } 0 \mid \text{Send } 1) = \epsilon_1$$

$$P(\text{Rec: } 1 \mid \text{Send } 1) = 1 - \epsilon_1$$

$$P(\text{Rec } 0) = ? \quad P(\text{Rec} = 1) = ?$$

Assumption: $P(\text{Send } 0) = 1 - p$
 (known) $P(\text{Send } 1) = p$

Now let's compute

$$\begin{aligned} & P(\text{Error in Receiver}) \\ &= P(\{(\text{Rec } 0, \text{Send } 1), (\text{Rec } 1, \text{Send } 0) \}) \\ \text{Total probability} &= \underbrace{P(\text{Error} | \text{Send } 0)}_{B_1} \cdot \underbrace{P(\text{Send } 0)}_{B_1} + \\ & \quad P(\text{Error} | \text{Send } 1) \cdot \underbrace{P(\text{Send } 1)}_{B_2} \\ &= \epsilon_0 \cdot (1 - p) + \epsilon_1 \cdot p \\ &= \epsilon_0 (1 - p) + \epsilon_1 p. \end{aligned}$$

- An important application of conditional probability is Bayes Rule. As before, assume we have a partition $\{B_1, B_2, \dots, B_n\}$ of the sample space S . Now we observe an event A occurs.

And we want determine the probability of $B_1, B_2, \dots, B_n$ given Event A :

$$\begin{aligned} P(B_i|A) &= \frac{P(A \cap B_i)}{P(A)} \\ &= \frac{P(A|B_i) P(B_i)}{P(A)} \\ &\stackrel{\text{Total Probability}}{\downarrow} = \frac{P(A|B_i) P(B_i)}{\sum_{k=1}^n P(A|B_k) P(B_k)} \end{aligned}$$

$P(B_i), i=1, 2, \dots, n$ are called the **prior** or **a prior distribution** of the events. Early

$P(B_i|A), i=1, 2, \dots, n$ are called the **posterior** or **a posterior distribution** of the events. Late/After

e.g. We update our belief after observing an output of the state.

Q: How to use Bayes Rule?

Example: In communication systems, we often want to answer questions like

"What is Probability Symbol '1' was sent given that Symbol '1' was received?"

$$P(\underbrace{\text{Send 1}}_A \mid \underbrace{\text{Rec 1}}_B)$$

This is the reverse of the conditional probability.

Bayes rule

$$= \frac{P(\text{Rec 1} \mid \text{Send 1}) P(\text{send 1})}{P(\text{Rec 1})}$$

$$= \frac{P(\text{Rec 1} \mid \text{Send 1}) P(\text{Send 1})}{P(\text{Rec 1} \mid \text{Send 1}) P(\text{Send 1}) + P(\text{Rec 1} \mid \text{send 0}) P(\text{Send 0})}$$

$$= \frac{(1-\epsilon_1) P}{\epsilon_0(1-P) + (1-\epsilon_1) P}$$